P04: Makers Makin' It, Act II -- The Seequel

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PROJECT NAME: Fitness Genie
DESCRIPTION: Our website is a meal calorie calculator. Users can sign up/login, track a meal's calories through a db of known food calories, save them to their account, and view them later on a personal profile page. Users can look through premade graphs to determine how their meals are different

TARGET SHIP DATE: TBD

Program Components:

1) Flask App (Python)
    a) **__init__.py** (main app + routes)
        i) Creates Flask app, config, and session handling
        ii) Renders templates and connects frontend to backend logic
        iii) Routes:
            (1) **/register** adds a new user to the **users** table (**data.py**). Checks if the username is unique, hashes the password, stores it in session, then redirects to **/dashboard**.
            (2) **/login** checks users (**data.py**), verifies password hash, stores session, redirects to **/dashboard**.
            (3) **/logout** clears session, redirects to **/login**.
            (4) **/home** displays data (somehow) on the nutrition of different foods as well as another dataset of general calorie intake of a sample of 2000 people
            (5) **/calculate** allows users to select food that they eat in a day and it will calculate the calorie intake and compare it with the rest of the general data from the nutrition intake dataset by redirecting the to **/visualize**
            (6) **/visualize** Displays interactive graphs using D3.js based on:
                - User's calorie intake compared with general nutrition dataset
                - Macronutrient breakdown (carbs, protein, fat)
                Users can filter by:

- Nutrient type, etc.

(7) **/profile/** uses **data.py** to load a creator's saved meals, calories, and nutrition stats

b) **data.py**

i) Connects to SQLite3 database and creates/maintains tables:

(1) **users:** stores user account information.

(2) **foods:** food name, calories, nutrients

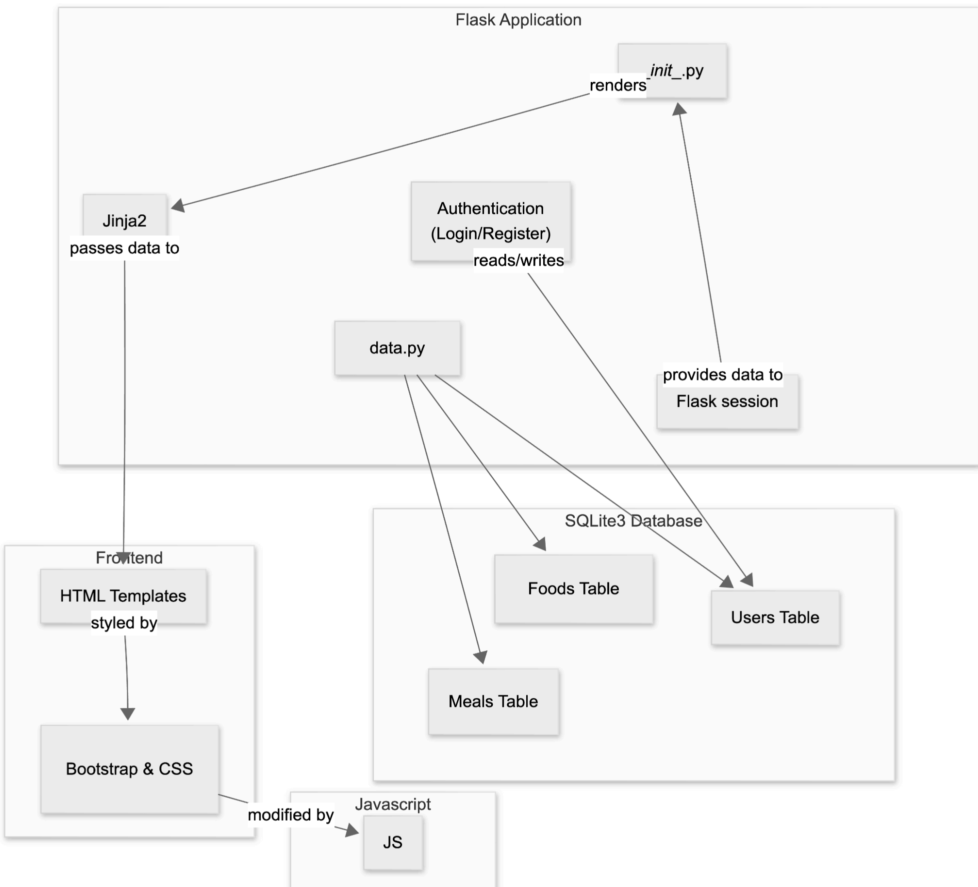
(3) **meals:** list of foods + user?

2) Templates (HTML)
   a) **/login** (username + password. Error message if login fails).
   b) **/register** (username + password. Error message if username is taken).
   c) **(home.html) /home** (Displays: Overview of nutrition data, basic D3 visualizations, Info about healthy calorie ranges)
   d) (**calculate.html**) **/calculate** Allows users to:
      - Select foods from database
      - Input quantity
      - Calculate total daily calorie intake
      - Compare result with general dataset
   e) (**visualize.html**) **/visualize** Interactive D3 graphs:
      - Line chart (calories over time or vs. average)
      - Bar chart (macronutrient breakdown)
      - Comparison chart (user vs dataset)
   f) (**profile.html**) **/profile/<username>** (shows username, nutritional stats).
   g) **error.html** (invalid id's or permission errors).
3) Database (SQLite3) (stored in data.db)
   a) **users** table stores all usernames and password hashes for authentication.
   b) **foods** nutritional data
   c) **meals** user meal logs
4) Frontend Framework
   a) **Bootstrap**: used for navbar, forms (login/registration), buttons, layout/grid, and cards for food items to keep styling consistent and responsive.
5) JavaScript
   a) Dynamic food selection on /calculate
   b) Updating totals in real time
   c) Rendering D3 visualizations
   d) Sends the updated tier list state to flask via POST so **data.py** can update the database.
6) RESTful APIs

a) None

7) External CSS (if needed)

Component Map:



Database Organization

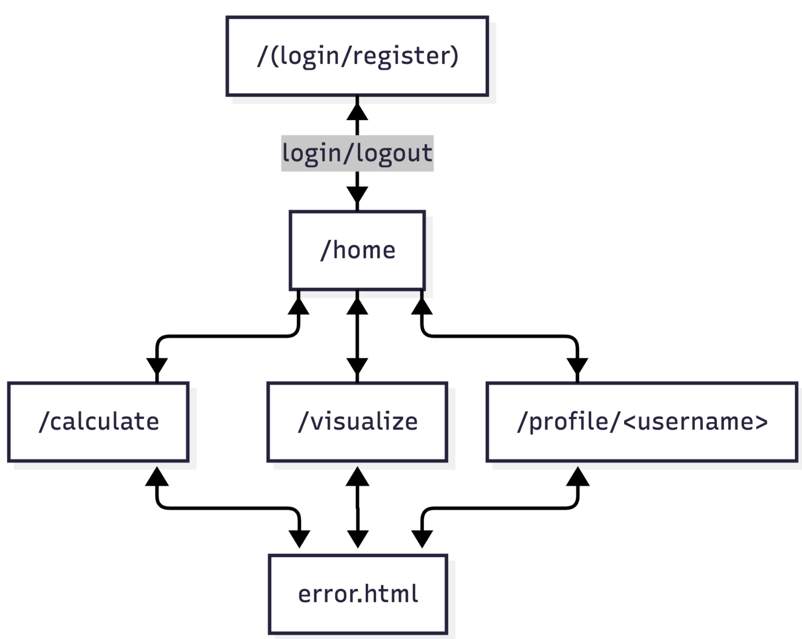
USERS			
TEXT	name	PK	Unique
TEXT	password		For authentication
TIMESTAMP	created_at		DEFAULT CURRENT_TIMESTAMP

Foods

INTEGER	id	PK	Auto-increment
TEXT	name		
INTEGER	calories		
FLOAT	protein		(grams)
FLOAT	carbs		(grams)
FLOAT	fat		(grams)

Meals			
INTEGER	id	PK	Auto-increment
TEXT	username	FK	USERS(username)
INTEGER	food_id	FK	FOODS(id)
FLOAT	quantity		

Site Map:



Visualization Library

D3 JS

APIs

(None needed)

Datasets

- https://www.kaggle.com/datasets/abdussamad123/user-daily-nutritional-intake
- https://www.kaggle.com/datasets/utsavdey1410/food-nutrition-dataset/data

Front End Framework

Bootstrap

Tasks & Assignments

TASK BREAKDOWN

Task	**Devo(s)**
Login & Register	James Sun
Editor UI & Javascript	Matthew Ciu
DB Organization	Yuhang Pan
D3JS Visualizations	Thomas Mackey